

Problem 1.

The correct answer is A.

Problem 2.

The correct answer is C.

Problem 3.

$$\text{a) } \omega_1 = \sqrt{\frac{3}{2} \frac{g}{L}}$$

$$\text{b) } v_{\text{cm}} = \sqrt{\frac{3}{2} \cos \theta \, gL}$$

Problem 4.

The correct answer is B and E.

Problem 5.

$$\text{a) } \omega = \frac{2}{\sqrt{3}} \sqrt{\frac{FL}{MR^2}}$$

$$\text{b) } \omega = \frac{2\sqrt{2}}{\sqrt{3}} \sqrt{\frac{FL}{MR^2}}.$$

Problem 6.

$$\omega = \sqrt{\frac{mgh}{\left(\frac{1}{2}m + \frac{1}{4}M\right) R^2}}$$